

Big Data, Bigger Bags

Project 4 — Digital Analytics Mini-Deliverable

Project 5 — Pivot Table Analysis

Dataset: Google Merchandise Store Synthetic Dataset

November 2020 – January 2021

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1 Scenario Recap + Manager Question

Big Data, Bigger Bags operates a tech merchandise e-commerce store. During the November 2020–January 2021 holiday window, the store attracted significant traffic, yet purchase volumes remained disproportionately low relative to product views. Our manager, **Frankie Velasquez, Director of E-Commerce Growth**, has tasked our team with diagnosing where shoppers disengage and which product categories drive complete buyer journeys — findings that will directly shape the spring catalog refresh and website UX roadmap.

Decision	Business Question
Funnel Optimization	At which stage is drop-off most severe, and is it a product or UX issue?
Catalog Strategy	Which product categories browse well but fail to convert — and should they stay in the spring catalog?

2 KPI + Decision Connection

KPI	Formula	Data Source	Decision Supported
Stage Conversion Rate	$(N_{+1} \div N) \times 100$	funnel_stage	Pinpoint the weakest funnel link
Stage Drop-off Rate	100% – Conversion Rate	funnel_stage	Quantify revenue leakage by step
Category View-to-Buy Rate	$(\text{Purchases} \div \text{view_item}) \times 100$	item_category, event_name	Catalog inclusion / exclusion
Browse-but-No-Buy Index	Views (0 purchases) ÷ Total views	item_name, event_name	UX and placement decisions
Avg. Engagement Time	Mean engagement_time_ms	engagement_time_ms	Detect friction by stage

3 How the KPI Equations Were Derived

Stage Conversion Rate counts the raw events logged at the next funnel stage and divides by the current stage, then multiplies by 100 to express as a percentage. Each funnel stage is a discrete `event_name` field in the Funnel & Behavior sheet, so the “count” is simply a row count filtered per stage.

Stage Drop-off Rate is the complement: $100\% - \text{Conversion Rate}$. It answers “what share of users *did not* proceed?” using the same inputs.

Category View-to-Buy Rate filters the same sheet twice — once for `event_name = view_item` and once for `event_name = purchase` — groups both counts by `item_category`, then divides purchases by views per category. The result surfaces which categories turn browsers into buyers.

Browse-but-No-Buy Index identifies products where the purchase count is zero, sums their view events, and divides by total view events. A higher index means more of the store’s browse traffic leads nowhere — a direct flag for UX or catalog action.

Avg. Engagement Time averages the `engagement_time_ms` field grouped by `funnel_stage`. Longer dwell time at a stage can indicate genuine interest *or* friction — interpreted alongside drop-off rate to distinguish the two.

4 Analytics Snapshot

Figure 1 — Conversion Funnel: Event Volume by Stage

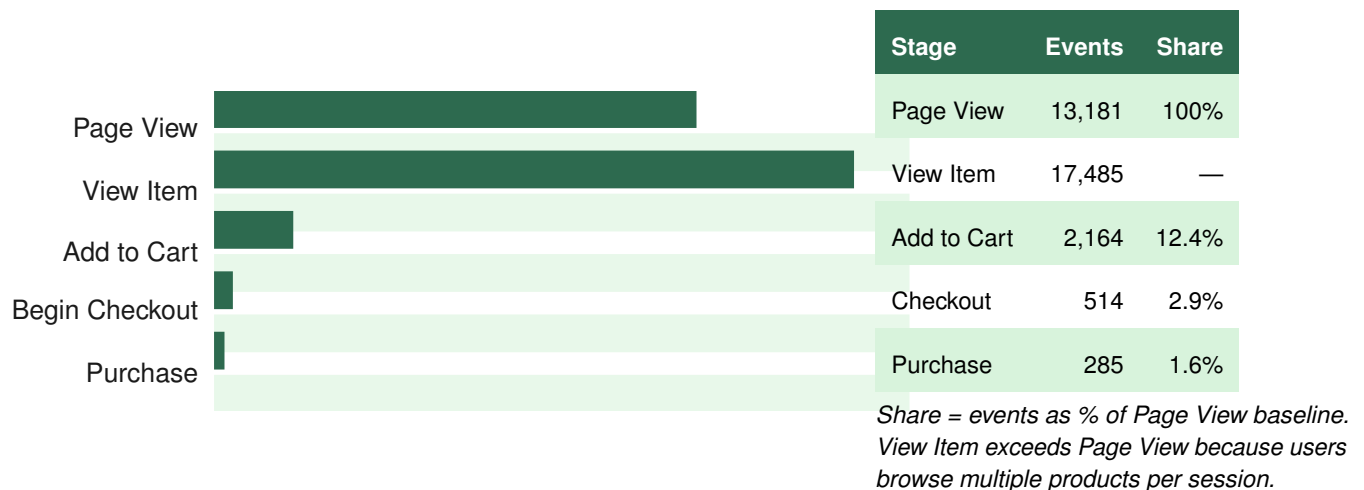
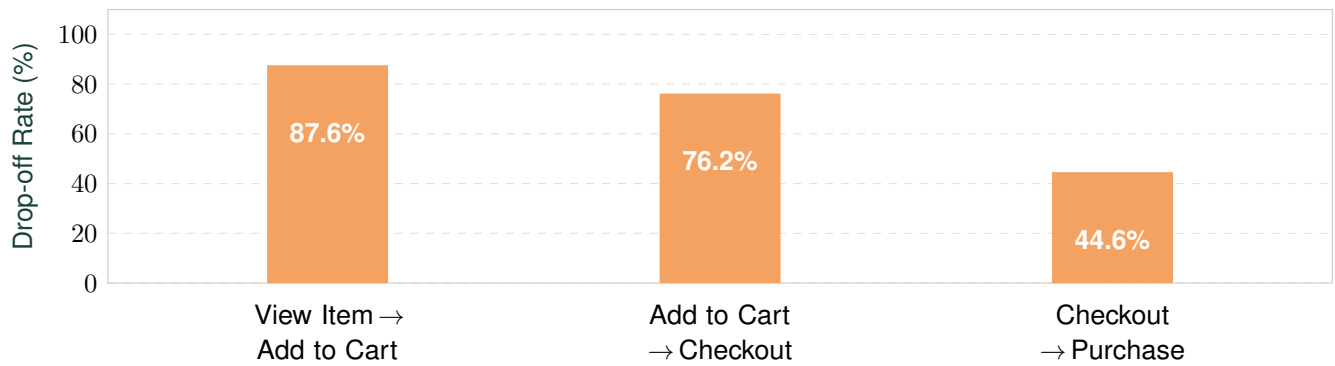
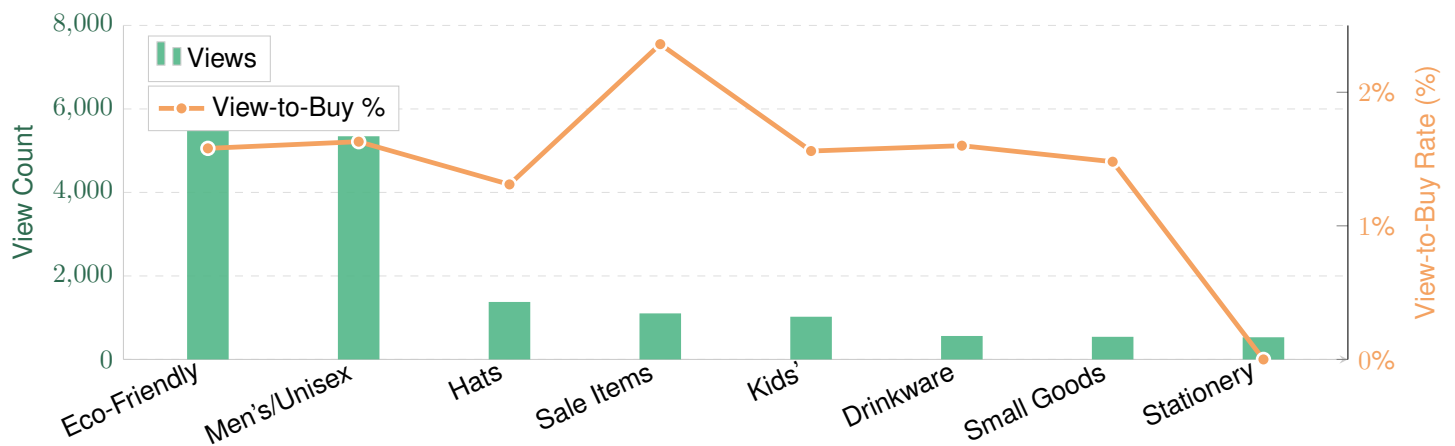


Figure 2 — Stage-to-Stage Drop-off Rate (%)



The 87.6% drop from View Item to Add to Cart is the single largest funnel leak. Even users who begin checkout abandon at a 44.6% rate — signalling a trust or friction barrier.

Figure 3 — Product Category: View Volume & View-to-Buy Conversion Rate



Green bars show raw view volume per category. Orange line shows the view-to-buy conversion rate on the right axis — making the purchase gap visible without tiny bars. Stationery converts at **0%**; Sale Items punch above their weight at **2.4%**.

Figure 4 — Top 10 Most-Viewed Products: Browse vs. Buy

Product Name	Views	Purchases	View→Buy %
Google Land & Sea Nalgene Water Bottle	622	9	1.4%
Android Jotter Task Pad	574	13	2.3%
Google Recycled Pen Blue	548	7	1.3%
Google Jotter Task Pad	547	13	2.4%
Google Land & Sea Journal Set	536	6	1.1%
Google Recycled Pen Black	536	13	2.4%
Google Land & Sea Women's Eco Tee	534	12	2.2%
Google Super G Tumbler (Blue Lid)	531	5	0.9%
Google Super G Tumbler (Red Lid)	516	4	0.8%
Google Land & Sea French Terry Sweatshirt	516	9	1.7%

Orange = View-to-Buy Rate below 1.5% — flagged for product page audit before spring launch.

5 Key Findings

- **The funnel bleeds most at the View-to-Cart stage (87.6% drop).** We should prioritize reducing friction at the product page level — not at checkout — since the problem starts much earlier in the journey.
- **Only 285 purchases completed from 17,485 product views** — a store-wide view-to-purchase rate of 1.6%. Improving this by even 0.5 points would meaningfully move revenue without increasing traffic spend.
- **Eco-Friendly and Men's/Unisex are the two strongest-performing categories** on both view volume and conversion rate. These should anchor the spring catalog and receive the most promotional investment.
- **Stationery/Stickers generated 531 views but zero purchases** — the worst browse-to-buy gap in the dataset. We recommend deciding whether to reposition or remove this category before the spring refresh.
- **Tumbler products convert below 1%** despite top-10 view volume. The likely causes are price sensitivity or weak product page quality — both fixable before launch.
- **The checkout-to-purchase step loses 44.6% of already-intent users.** This is a specific technical or trust barrier that should be investigated with session replay tools before any catalog changes are made.

6 Recommended Actions

1. **Audit product pages for top-viewed, low-converting items.** For Tumblers and Stationery/Stickers, we recommend commissioning a UX review covering descriptions, image quality, pricing display, and CTA button placement before the spring catalog is finalized.

2. **A/B test a social proof nudge on the top 10 viewed product pages.** Since 87.6% of viewers never add to cart, we recommend introducing a low-friction prompt (e.g., “X people bought this today” or a sticky CTA) to directly target the biggest drop-off point.
3. **Simplify the checkout flow and add trust signals.** The 44.6% drop between Begin Checkout and Purchase among intent-confirmed users points to friction or trust failure. We recommend reducing checkout steps and adding security badges or a guest-checkout option before the spring launch.

7 Data Limitations + Digital Risk

Type	Item	Detail
Limitation 1	Holiday-only time window	Data covers Nov 2020–Jan 2021, a peak shopping period. Funnel behavior may not reflect typical year-round patterns, risking over-investment in seasonal demand.
Limitation 2	Event counts, not unique users	Each row is an event, not a person. One user can generate multiple <code>view_item</code> events per session, inflating the View Item count and exaggerating the View→Cart drop-off at the individual level.
Risk 1	Misattributing drop-off to product vs. technical bugs	Acting on funnel data without ruling out technical errors (e.g., a broken “Add to Cart” button) could lead to catalog changes that miss the root cause.
Risk 2	Over-investing in holiday-driven top categories	Eco-Friendly and Men’s/Unisex lead during a peak window. Budget misallocation is possible if seasonal demand — not enduring preference — was the actual driver.
Mitigation 1	Validate with session recordings before catalog decisions	Use session replay tools (e.g., Hotjar, FullStory) to confirm drop-off reflects user behavior — not a UI or technical failure — before any catalog action.
Mitigation 2	Set a minimum conversion threshold for catalog inclusion	Require $\geq 1.5\%$ view-to-purchase rate across a non-holiday sample period before including any product in the spring refresh.

8 Team Roles + AI Use Statement

Generative AI tools were used by our team to support the structuring of written sections, refine KPI definitions for clarity and precision, and assist in drafting the managerial interpretation and risk sections. AI was also used to synthetically modify the Google Merchandise Store dataset: the original data was pulled from Google’s BigQuery public dataset, however Google does not disclose certain transaction-level values. AI was used to fill those gaps with synthetic values consistent with the dataset’s real distributions and structure. All findings are derived from this synthetically completed dataset. Representative prompts included: “*Refine this KPI formula for operational clarity*” and “*Suggest specific mitigations tied to funnel data governance.*” All AI outputs were reviewed, edited, and adapted by our team before inclusion.

1 Scenario Recap + Manager Decisions

Big Data, Bigger Bags operates a tech merchandise e-commerce store. Despite strong holiday traffic between November 2020 and January 2021, purchase volumes remained disproportionately low. Our manager, **Frankie Velasquez, Director of E-Commerce Growth**, needs a structured breakdown of revenue patterns, product performance, and funnel behavior to guide three upcoming decisions before the spring catalog launch.

Decision	What Frankie Needs to Know
Monthly Trend Review	How did funnel volume shift across Nov, Dec, and Jan — and where did momentum drop?
Catalog Prioritization	Which product categories convert well enough to anchor the spring catalog, and which should be cut or repositioned?
Product Page Investment	Which specific products attract the most views but fail to convert, making them the highest-ROI targets for a UX or content fix?

2 KPI + Business Question Setup

KPI	Formula	Source Tab	Decision Supported
Monthly Event Volume	Count of <code>event_name</code> by month and <code>funnel_stage</code>	PT1 Monthly Funnel Trend	Trend and seasonality review
Stage Drop-off Rate	$(1 - N_{+1}/N) \times 100$	PT3 Funnel Drop-off	Identify weakest funnel stage
Category View-to-Buy Rate	$(\text{Purchases} \div \text{Views}) \times 100$ per category	PT2 Category Performance	Catalog inclusion decisions
Product View-to-Buy Rate	$(\text{Purchases} \div \text{Views}) \times 100$ per product	PT4 Top Products	Product page audit priority
Browse-but-No-Buy Index	Views with 0 purchases $\div$ Total views	PT4 Top Products	UX and placement decisions
Funnel Completion Rate	$\text{Purchases} \div \text{Page Views} \times 100$	PT3 Funnel Drop-off	Overall store conversion health

3 Data Summary

Dataset: Google Merchandise Store Synthetic Dataset, sourced from Google's BigQuery public dataset and synthetically modified to fill transaction-level gaps Google does not publicly disclose.

Time Period: November 2020 – January 2021 (3-month holiday window)

Total Records: 33,629 events across 5 funnel stages

Key Fields Used: `event_name`, `funnel_stage`, `item_name`, `item_category`, `month`, `price`, `quantity`, `line_total`, `engagement_time_ms`, `is_view`, `is_purchase`

Dataset Limitations: The dataset covers only a holiday period, meaning behavior may skew higher than typical. Event-level data means one user can appear multiple times per session, inflating stage counts. Certain transaction values were synthetically generated to complete the dataset where Google's public data had gaps.

4 Pivot Table Analysis

PT1 — Monthly Funnel Trend

This pivot table shows event counts broken down by **month** (rows) and **funnel stage** (columns), giving Frankie a month-by-month view of how traffic moved through the funnel from November to January.

Why it matters: This is the trend view. It reveals that December was the peak month across all stages — driven by holiday shopping — while January saw a sharp drop in both views and purchases. This tells Frankie that the store is heavily holiday-dependent and that January demand fell off sharply, which should inform how aggressively the spring catalog is promoted heading into Q1.

PT2 — Category Performance

This pivot table compares **view_item** counts and **purchase** counts by `item_category`, with a calculated View-to-Buy Rate % column added manually. It shows which categories attract browser attention versus which ones actually close the sale.

Why it matters: This is the primary comparison view. Eco-Friendly and Men's/Unisex lead in both volume and conversion, confirming they belong in the spring catalog. Stationery/Stickers shows 531 views with zero purchases — the clearest signal in the dataset that this category either has a product page problem or simply does not resonate with buyers. Sale Items outperform on conversion rate (2.36%) relative to their view volume, suggesting promotional pricing is an effective lever.

PT3 — Funnel Drop-off

This pivot table counts total events per `funnel_stage` and includes a manually calculated drop-off rate column showing the percentage of users lost between each stage.

Why it matters: The 87.6% drop from View Item to Add to Cart is the single most actionable number in the

entire dataset. It means the store is losing nearly 9 out of every 10 potential buyers at the product page — before they even consider purchasing. The secondary drop of 44.6% from Begin Checkout to Purchase suggests that even users who commit to checkout face a friction or trust barrier.

PT4 — Top Products

This pivot table ranks the **top 10 most-viewed products** by view count (using the `is_view` helper column) alongside their purchase counts (`is_purchase`) and a calculated View-to-Buy Rate %.

Why it matters: High view counts with low conversion rates flag specific products where the investment in a product page fix would have the highest return. Google Super G Tumbler products rank in the top 10 for views but convert below 1%, making them the clearest candidates for a description, pricing, or image quality review before the spring launch.

5 Charts and Visual Summary

Chart	Type	Source	What It Shows and Why It Matters
Chart 1	Clustered Bar	PT3 Funnel Drop-off	Displays event volume at each of the 5 funnel stages. The visual immediately communicates the massive gap between View Item (17,485) and Add to Cart (2,164), making the 87.6% drop-off intuitive for any stakeholder. Frankie can use this to justify UX investment at the product page level.
Chart 2	Line Chart	PT1 Monthly Funnel Trend	Plots each funnel stage as a separate line across Nov, Dec, and Jan. The lines show a clear December peak followed by a January decline across all stages. This helps Frankie understand that holiday timing is a major driver of store performance and that Q1 will require proactive promotional support to sustain momentum.
Chart 3	Combo (Bar + Line)	PT2 Category Performance	Uses green bars for view volume on the primary axis and an orange line for View-to-Buy Rate % on a secondary axis. This dual-axis design makes conversion rate differences visible and comparable across all categories simultaneously, solving the problem of purchase counts being too small to display as bars next to view counts in the thousands.

6 Findings and Recommendations

Key Findings:

- **December was the peak month across all funnel stages.** View Item events hit 7,218 in December versus 3,135 in January — a 56% drop. The store is highly seasonal and Frankie must build a spring campaign strategy that does not rely on organic holiday demand.
- **The funnel loses 87.6% of users between View Item and Add to Cart.** This is the largest single drop in the entire funnel and occurs before checkout — meaning the problem is at the product page, not the payment flow.
- **Only 1.6% of all product views result in a purchase.** Across 17,485 view events, only 285 purchases were completed. A 0.5 percentage point improvement would represent meaningful revenue gain without any increase in traffic spend.
- **Eco-Friendly and Men's/Unisex are the two strongest categories** by both view volume and conversion rate. These should receive the most prominent placement and promotional budget in the spring catalog.
- **Sale Items convert at 2.36%** — the highest rate of any category — despite moderate view volume. Promotional pricing is an effective conversion lever that Frankie could apply more broadly.
- **Stationery/Stickers generated 531 views but zero purchases.** This is the clearest underperformer in the dataset — a candidate for removal or complete repositioning before the spring refresh.
- **Tumbler products rank top 10 in views but convert below 1%.** Google Super G Tumblers both convert under 1%, suggesting a disconnect between product appeal and purchase intent — likely driven by price point or weak product page content.
- **Jotter and Recycled Pen products consistently convert at 2.3–2.4%,** outperforming most of the catalog. These should anchor product page best-practice templates for the spring refresh.

Recommended Actions:

1. **Prioritize Eco-Friendly and Men's/Unisex for spring catalog anchor placement.** These two categories deliver on both volume and conversion. We recommend they receive the largest share of homepage real estate, email campaign slots, and paid promotion budget for the spring launch.
2. **Conduct a product page audit on Tumbler products and Stationery/Stickers before the spring refresh.** We recommend reviewing product descriptions, photography, pricing display, and CTA placement — using Jotter product pages as the internal benchmark for what a high-converting page looks like.
3. **Build a January recovery campaign to counter the post-holiday drop-off.** The PT1 trend data shows a sharp January decline across all funnel stages. We recommend Frankie's team plan a January promotion timed to land in early January to prevent the revenue cliff seen in this dataset from repeating.

7 Data Integrity + Responsible Interpretation

Type	Item	Detail
Limitation 1	Holiday-only window	All three months fall within the holiday shopping season. Conversion rates, category rankings, and funnel volumes will likely look different during non-holiday periods. Decisions based solely on this data risk over-indexing on seasonal behavior.
Limitation 2	Synthetic data gaps	Certain transaction-level values were not available in Google's public BigQuery dataset and were synthetically generated to complete the analysis. While synthetic values were designed to match real distributions, they should not be treated as ground-truth transaction records.
Risk 1	Treating event counts as unique user counts	Each row is a single event, not a unique person. One user can contribute multiple view events per session. Pivot tables counting events may overstate the number of individual shoppers at each stage.
Risk 2	Over-relying on a single season to set catalog strategy	Using only November–January data to decide the spring catalog could lead to misaligned investment. Categories that peak during holiday gifting may underperform in spring without the same gifting motivation driving demand.
Mitigation 1	Validate pivot findings with session-level data	Before making catalog or UX decisions, Frankie's team should validate event-level pivot findings using session-level analytics (e.g., Google Analytics 4 user reports or session replay tools) to confirm patterns hold at the individual user level.
Mitigation 2	Set a non-holiday performance benchmark before spring launch	Any category included in the spring catalog should meet a minimum View-to-Buy Rate of $\geq 1.5\%$ measured across a non-holiday data sample — not just the Nov–Jan window — before receiving promotional investment.

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